SESSION CHECK

Session 2 — First night: eyes only

Practical Astronomy Intensive

Name ______________________________ Date ________________

9 questions **20** minutes

The first observing night, checked partly at the eyepiece-less sky itself: two of these are practical tasks, because averted vision and tracing the Milky Way are things you watch a student do, not things you read about their doing. Dark adaptation is asked twice — it is the night's discipline, and one white phone screen undoes it.

*Circle one letter for each multiple-choice question. Write short answers on the lines. Tasks marked **Practical** are done, not written — your assessor will watch.*

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1 Standing outside, you can see steely-blue Vega, golden Arcturus and red Betelgeuse without any instrument. What does a star's colour tell you?

- a Its surface temperature, directly — blue Vega is genuinely hotter than red Betelgeuse
- b Nothing real — all stars are alike, and colour is just an effect of how far away they are
- c Its temperature, with red the hottest and blue the coolest
- d How near it is — the reddest stars are the closest, which is why their colour shows

2 A student looks at the Orion Nebula and complains that it is grey, nothing like the red and blue photograph he has seen. What is the honest explanation?

- a His eyes work in the dark just as in daylight, only dimmer, so he needs to look harder
- b At that light level only his rods are firing, and rods are colour-blind — the photograph stacked photons for an hour, which no eye can do
- c The telescope's optics are poor — a better instrument would show the colours
- d He should stare directly at the nebula's centre, where his vision is sharpest, and the colour will emerge

3 Twenty-five minutes into the dark, a student pulls out his phone to check the time. The screen is white. What has he just done?

- a Lost a little sensitivity, which will come back within a minute or two outside
- b Nothing to his eyes — dark adaptation is about learning to concentrate, not about the retina
- c Nothing much — one quick glance at a screen is harmless
- d Bleached his rods and handed back most of the twenty-five minutes — rhodopsin has to rebuild from scratch, which is exactly why torches are red

4 You are shown a faint object (M33, or a dim cluster member the assessor names). Find it and demonstrate averted vision, describing aloud what changes.

Practical — performed for your assessor.

5. Someone points at the Big Dipper (Saptarishi) and calls it a constellation. What is the correct picture?

   a It is an asterism, and since asterisms are unofficial they are of little real use for finding your way around

   b It is an asterism — the bright core of Ursa Major, which is a far larger patch of sky — and asterisms are exactly what observers navigate by

   c It is an asterism, meaning its seven stars are a genuine group close together in space

   d They are right — the Big Dipper is one of the 88 official constellations

6. A student's cousin has been given a certificate from a 'star registry' naming a star after her. What should you tell her?

   a Only the IAU names stars — the certificate is a souvenir and that name appears in no catalogue any astronomer uses

   b The name is official once registered, though it takes some years to appear in catalogues

   c It is fine as long as the star was an alpha star, since alpha is always the brightest of its constellation and therefore properly catalogued

   d It cannot be right, because each star has exactly one true name and that one is already taken

7. A student holds his planisphere flat like a road map, facing south, and cannot make anything match. Later at the eyepiece he hops confidently in the wrong direction. What has gone wrong in each case?

   a A sky map only reads correctly when turned to face the direction you are looking, horizon edge down — and at the eyepiece a diagonal flips the view left for right while a Newtonian turns it round

   b He should hold the chart fixed as he did; the eyepiece view matches the chart exactly, so the hop must have been mis-counted

   c Nothing is wrong with the chart handling — the chart's stars are simply plotted for a different date

   d Both problems disappear with a planetarium app, which orients itself and makes paper charts unnecessary

8. A student in a Bortle 8 city backyard wants to buy a light-pollution filter so she can see galaxies from home. What is the honest advice?

   a It will not work for galaxies — a broadband filter rejects a few lamp wavelengths and does nothing for broad-spectrum targets; that money is better spent on the drive to a dark site

   b There is nothing to be done — a bright city sky is simply how cities are

   c It is worth it, though light pollution is really only an inconvenience for astronomers anyway

   d Buy it — a light-pollution filter effectively restores a Bortle 8 sky for any target

9. Trace the Milky Way band across the sky with your arm, name the constellation the brightest part runs through, then put binoculars on it and tell the group what you see.

   Practical — performed for your assessor.